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IT 112 Web Systems and Technologies

BSIT-2B 2nd Semester

Crumbelle Ordering System

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Introduction

The Crumbelle Ordering System is our proposed way of making restaurant ordering easier, faster, and more organized. We noticed that in a busy restaurant setup, taking orders manually can sometimes cause delays and mistakes, which can affect both customers and staff. With this system, we want to make things more convenient by letting customers easily browse the menu, place their orders, and track them without hassle. At the same time, it helps our staff manage orders, update menu items, and keep track of sales in a smoother and more efficient way. This system is our simple solution to improve service, reduce errors, and make the dining experience better for everyone at Crumbelle.

System Purpose

The purpose of the Crumbelle system is to make ordering faster, easier, and more organized for both customers and staff. It helps reduce errors, improve customer service, and manage orders and transactions more efficiently. The system also aims to provide a convenient and modern way of handling restaurant operations.

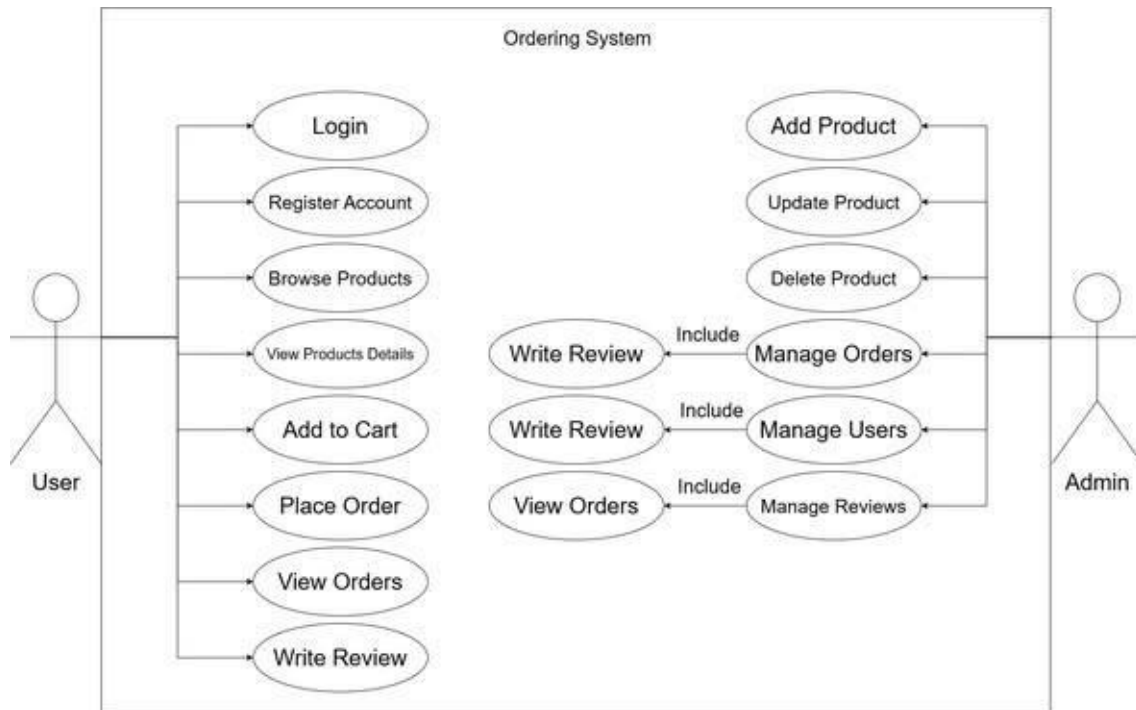
Target Users

- Customers
- Admin (Store Manager)

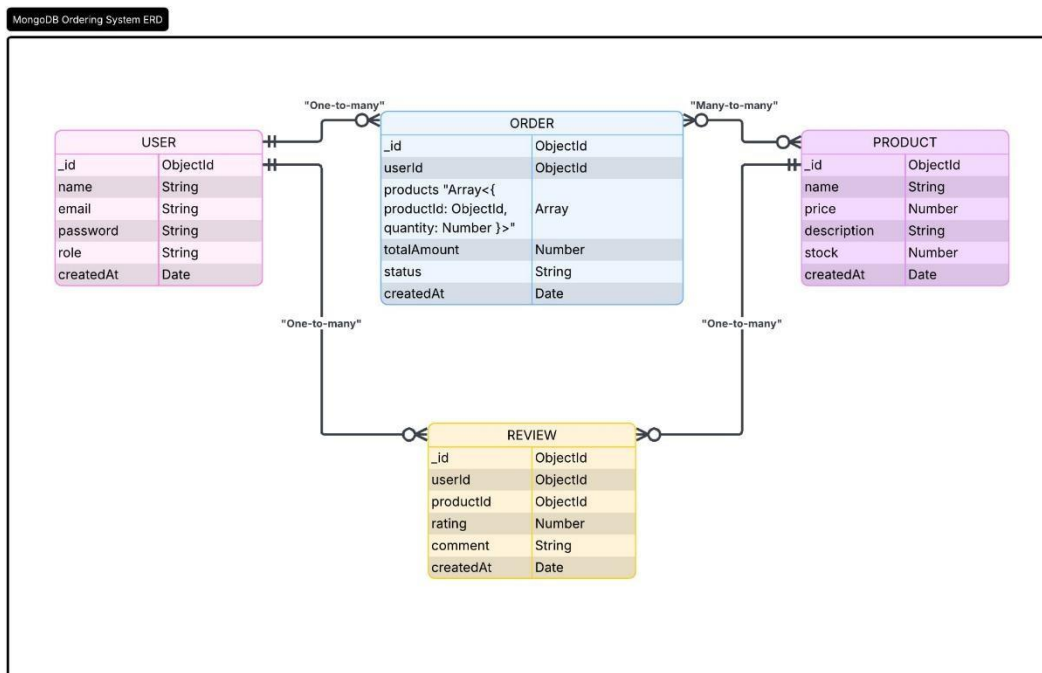
Key Features:

1. User Registration and Login (JWT Authentication)
2. Browse Products and View Details
3. Add to Cart and Place Orders
4. Order Management (Admin side)
5. Responsive User Interface

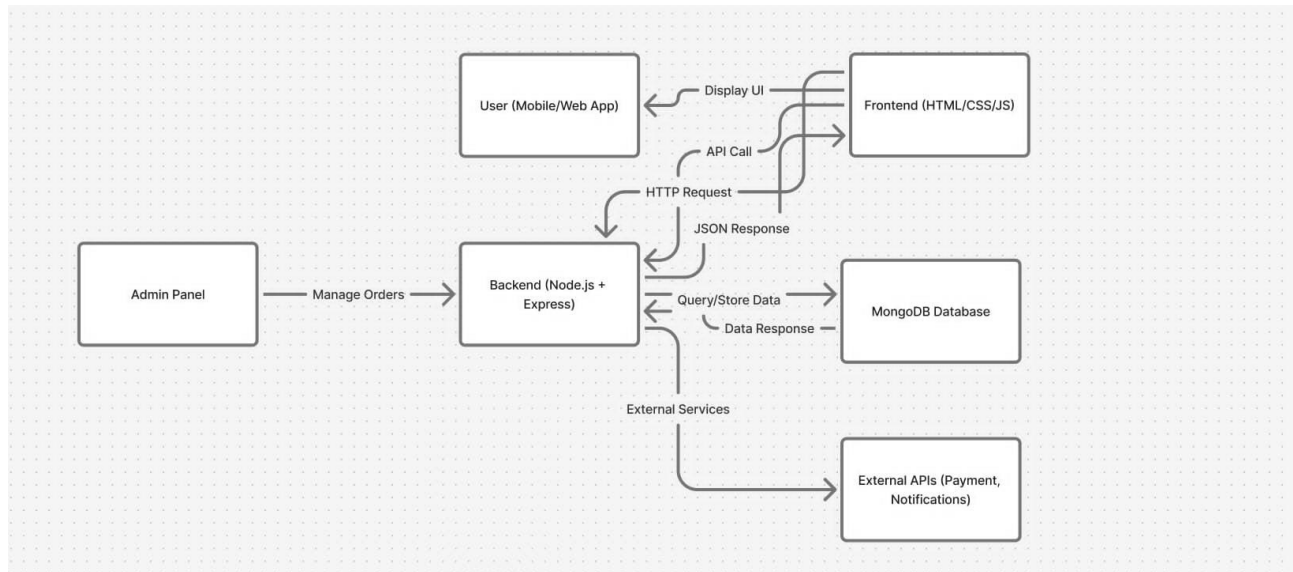
Use Case Diagram



Entity-Relationship Diagram



System Architecture Diagram



System Flow Explanation

- The user accesses the system through a web browser.
- The frontend (HTML, CSS, JavaScript) displays the user interface, allowing the user to browse products and interact with the system.
- When the user performs an action (e.g., placing an order), the frontend sends an HTTP request/API call to the backend.
- The backend (Node.js + Express) receives the request and processes it using appropriate API routes.
- The backend validates the request and performs necessary operations such as handling orders or retrieving product data.
- If needed, the backend communicates with the MongoDB database to store or retrieve data.
- The database processes the query and sends a data response back to the backend.
- The backend may also send requests to external APIs (e.g., payment or notification services) for additional functionality.
- The backend then sends a JSON response back to the frontend.
- The frontend updates the user interface and displays the result to the user (e.g., order confirmation).